

MIT

Academy of
Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

AI-Driven Early Detection and Classification of Water Quality for Safe Drinking Water

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Presentation Overview

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Problem Statement :

Project aims to use Machine Learning to predict whether water is safe for drinking based on its physical and chemical parameters like pH, hardness, and turbidity.

It helps in early detection of water contamination to ensure clean and safe water for all.

- Predict water potability using Machine Learning
- Dataset: 3276 samples from Kaggle
- 9 chemical and physical parameters used
- Final outcome → Safe / Not Safe prediction



Project Proposal:

Problem to Solve

- The main problem is that people do not have an instant and affordable way to check whether drinking water is safe or unsafe. Laboratory testing is slow, costly, and not easily accessible.

Dataset Source

- The dataset is taken from Kaggle – Water Potability Dataset
- It contains 3276 rows and 9 water quality features, along with a target column “Potability”.

Expected Outcomes

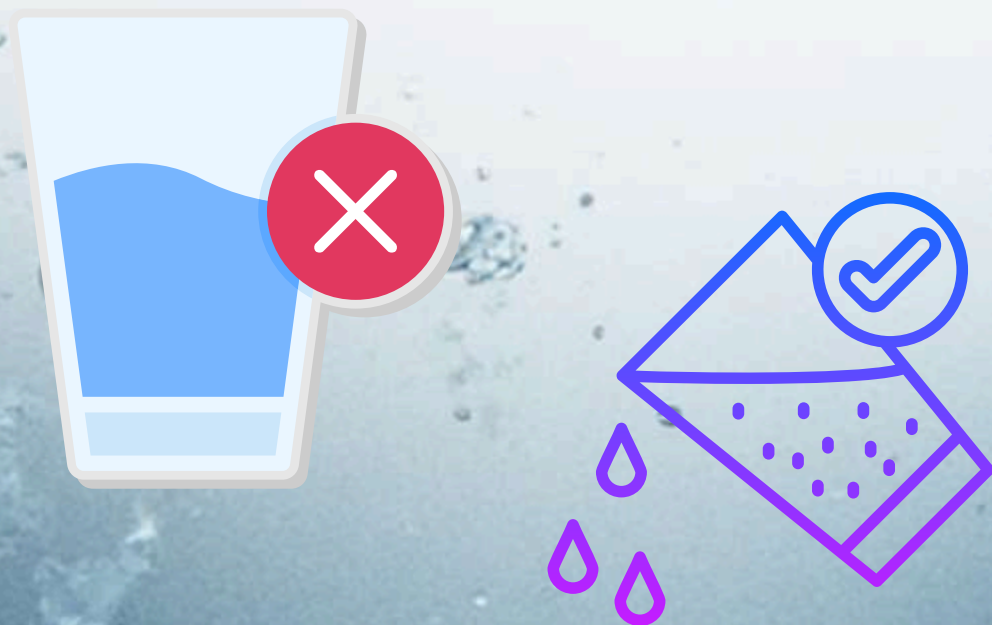
- Build a Machine Learning model that can predict Safe / Not Safe water
- Compare multiple ML algorithms and select the best-performing model
- Analyze feature importance and understand which parameters affect water safety
- Develop a user-friendly web application for real-time predictions

Motivation

- Ensuring clean and safe drinking water is essential for public health.
- This project aims to create a fast, intelligent, and accessible solution that helps users instantly check water safety without laboratory equipment.

Objectives :

- 1 To Classify water samples as potable (safe) or non-potable (unsafe).
- 2 To Support environmental safety and health monitoring.



SDG Goal :

SDG 6 – Clean Water and Sanitation

- Ensuring availability and sustainable management of clean water for all.



Dataset Description

- **Source :**

Kaggle Water Potability Dataset

- **Features Used:**

pH, Hardness, Solids, Chloramines, Sulfate,
Conductivity, Organic Carbon, Trihalomethanes,
Turbidity

- **Target Variable:**

Potability (1 = Safe, 0 = Unsafe)

- **Preprocessing Steps:**

- 1) Handling missing values
- 2) Normalization/scaling
- 3) Feature selections.

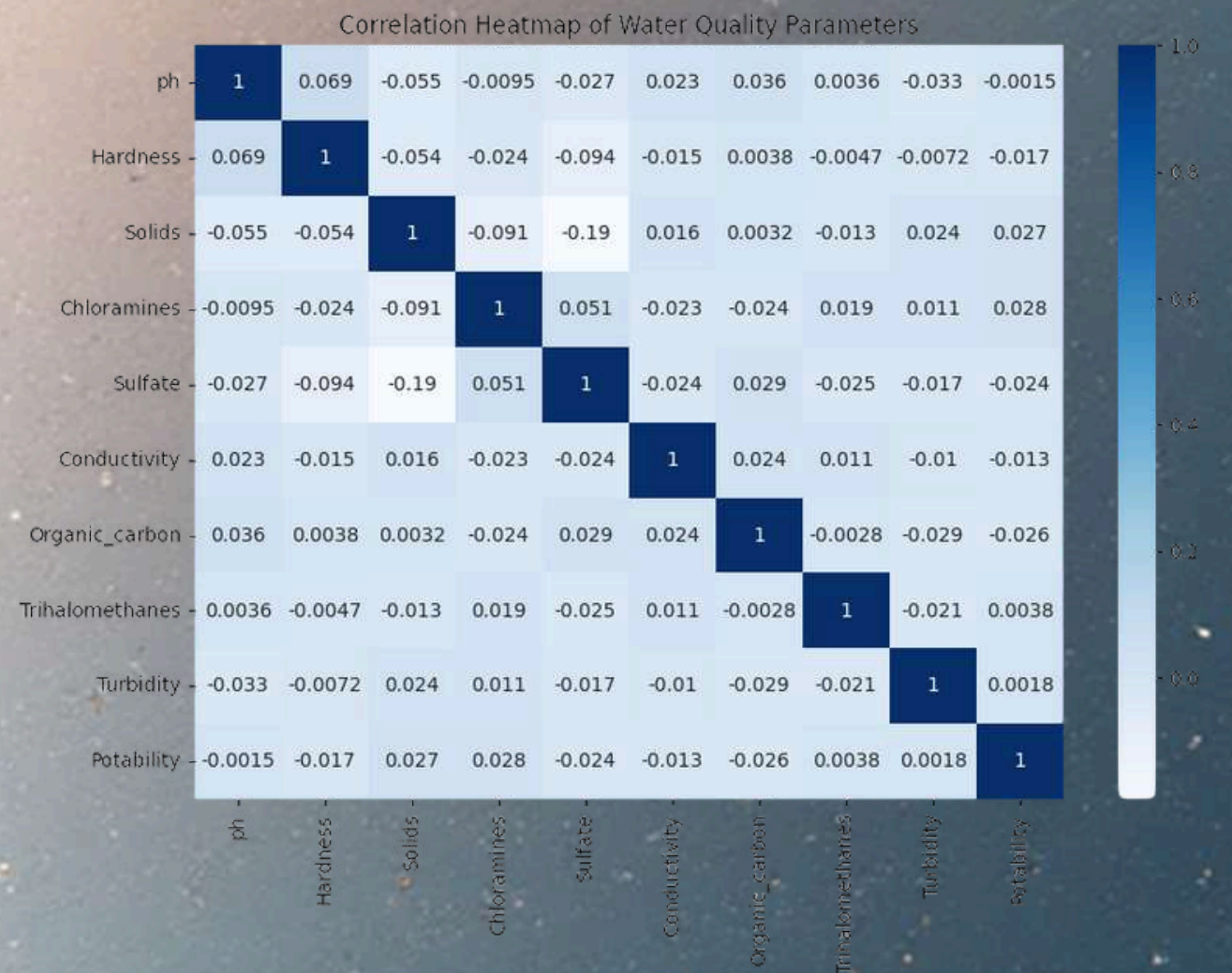
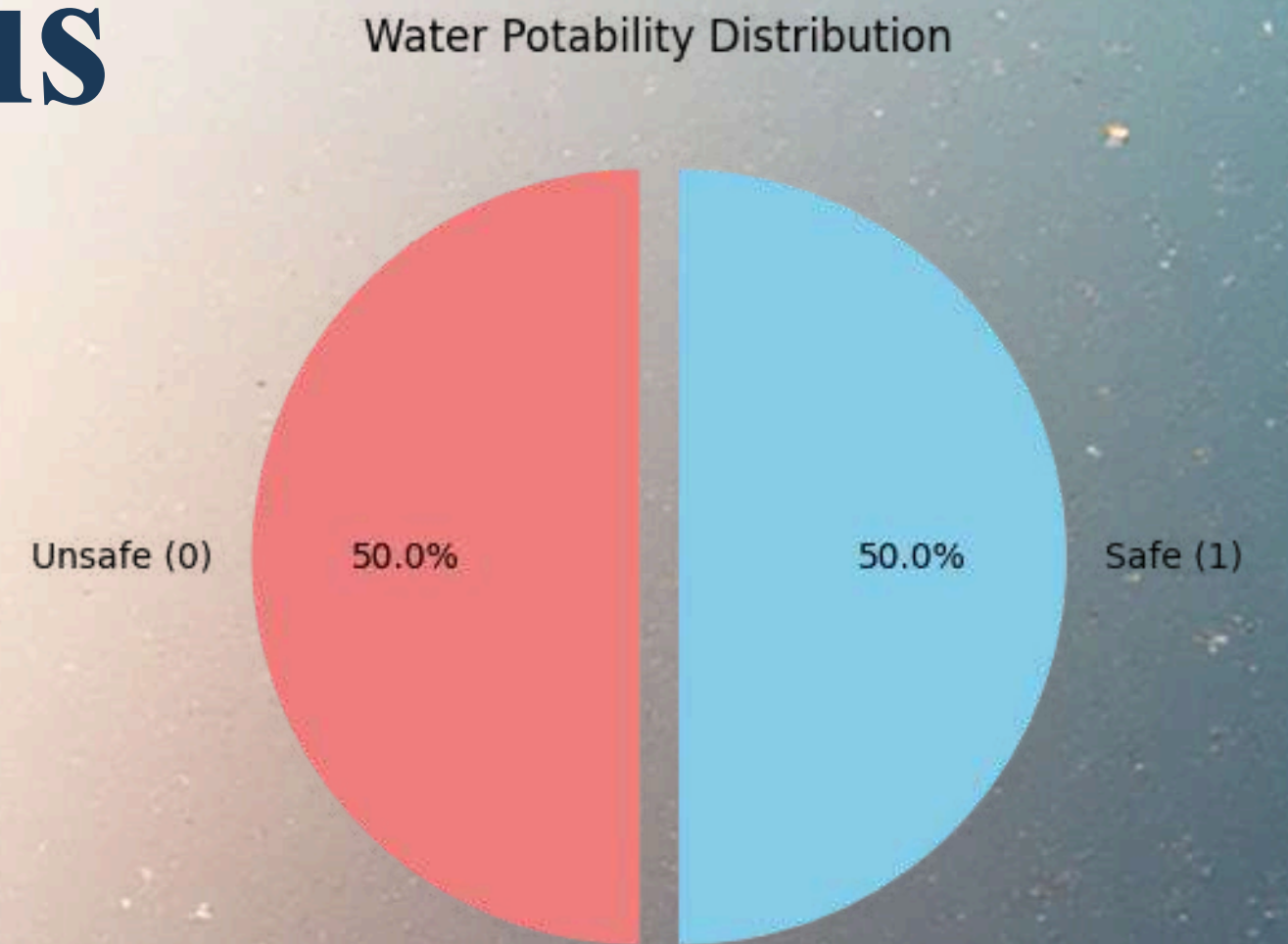
Dataset Link



Exploratory Data Analysis (EDA)

Key Points :

- Dataset contains 9 features and 1 target (Potability).
- Checked and handled missing values in features like pH, Sulfate, and Trihalomethanes.
- Data shows slight imbalance between potable (1) and non-potable (0) samples.
- Applied scaling/normalization for uniform feature range.
- Found correlation between Solids and Conductivity.



Flow Chart Creation Process

Data Collection



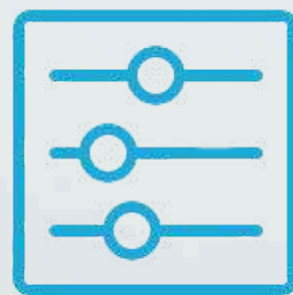
Model Training



Deployment



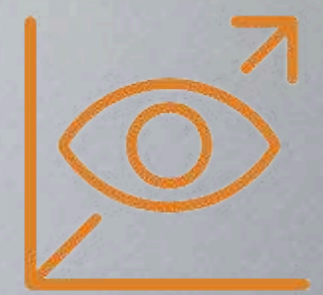
Preprocessing



Evaluation



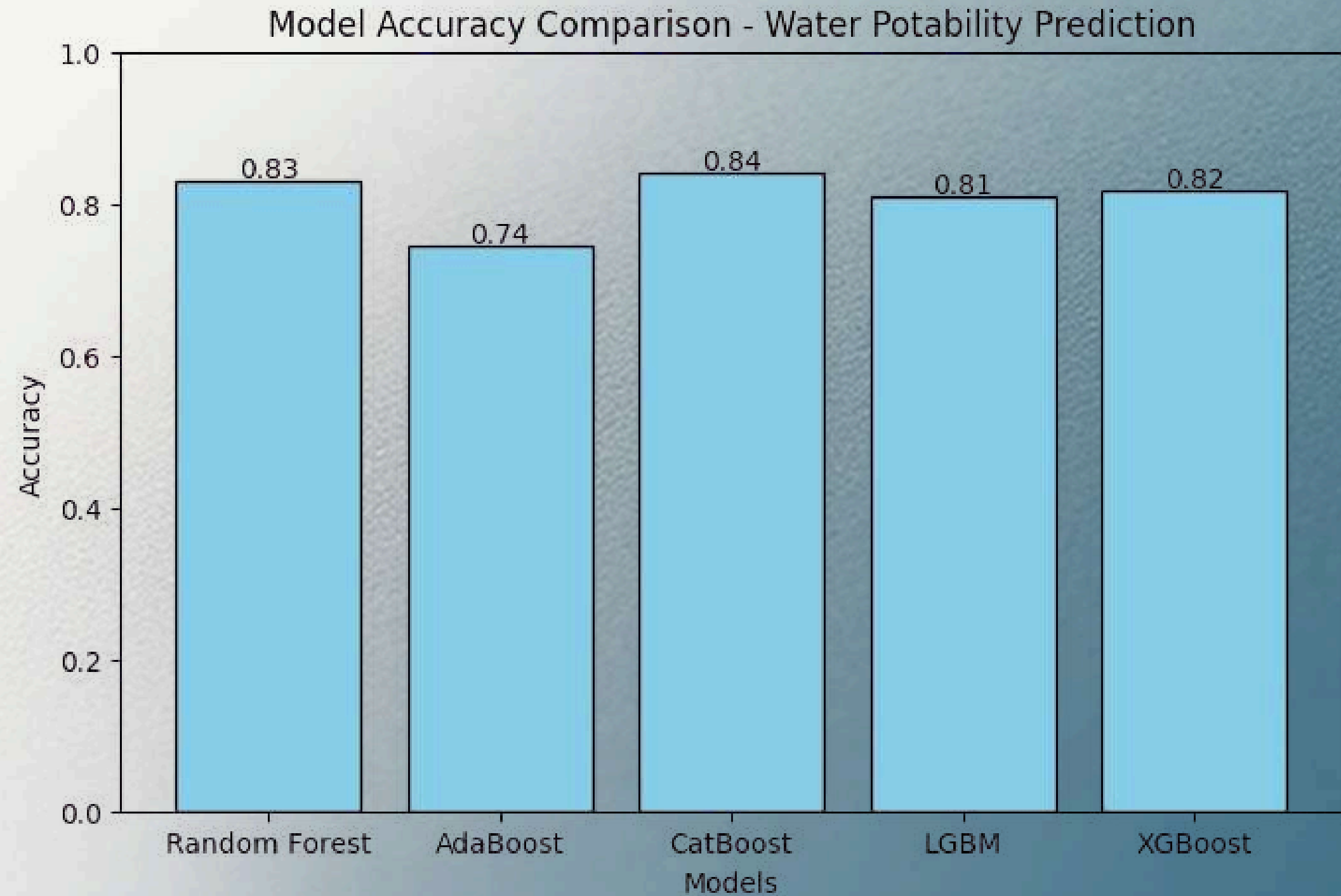
Monitoring



Model Selection Criteria

ML Models Compared:

- 1) Random Forest
 - 2) XGBoost
 - 3) CatBoost
 - 4) LGBM
 - 5) AdaBoost
- Selection based on performance metrics:
 - Accuracy, Precision, Recall, F1 Score
 - Best model chosen after comparative analysis.

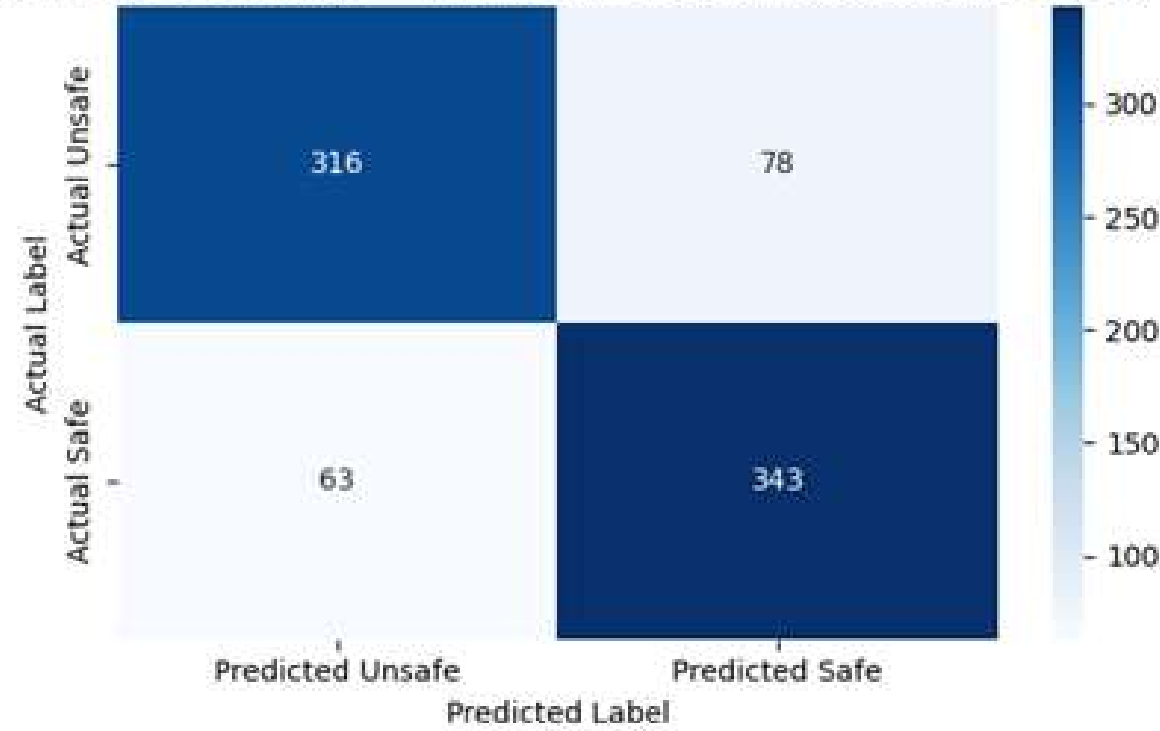


Model Comparison Table :

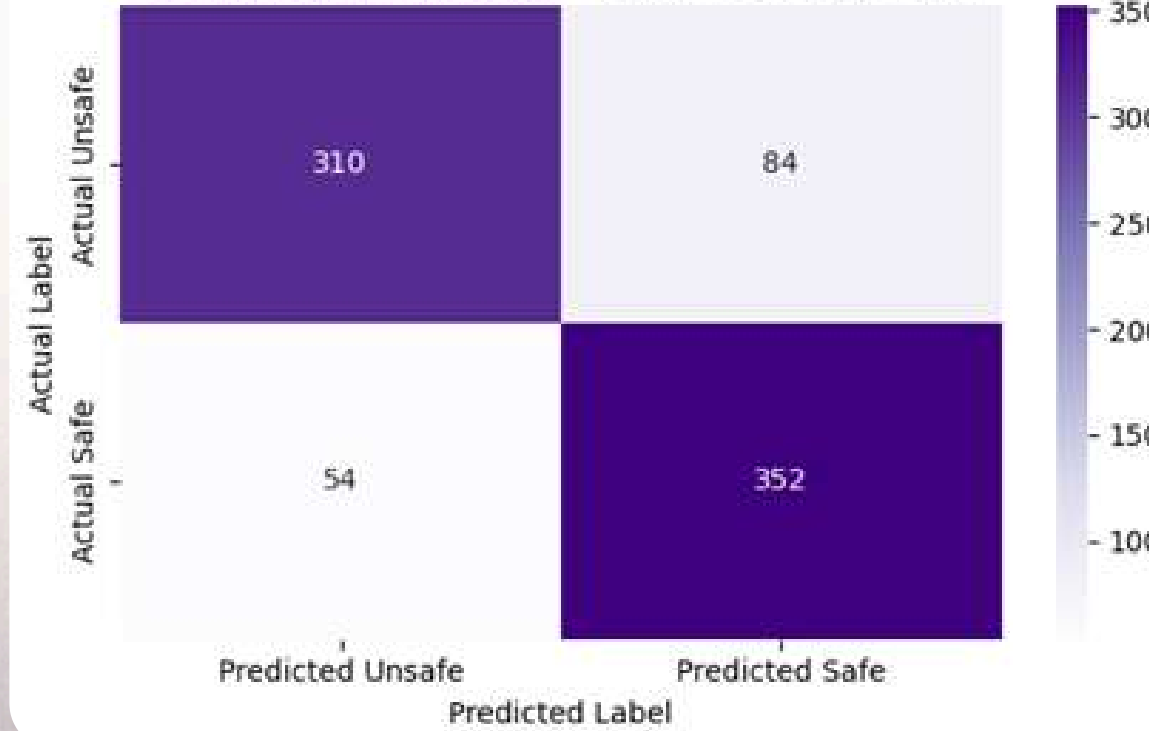
Model	Accuracy	Precision	Recall	F1 Score	R ² Score
Random Forest	0.83	0.812	0.865	0.838	0.32
AdaBoost	0.745	0.742	0.764	0.752	-0.02
CatBoost	0.84	0.82	0.877	0.848	0.36
LGBM	0.809	0.792	0.845	0.818	0.235
XGBoost	0.816	0.796	0.857	0.826	0.265

Visualization of Results :

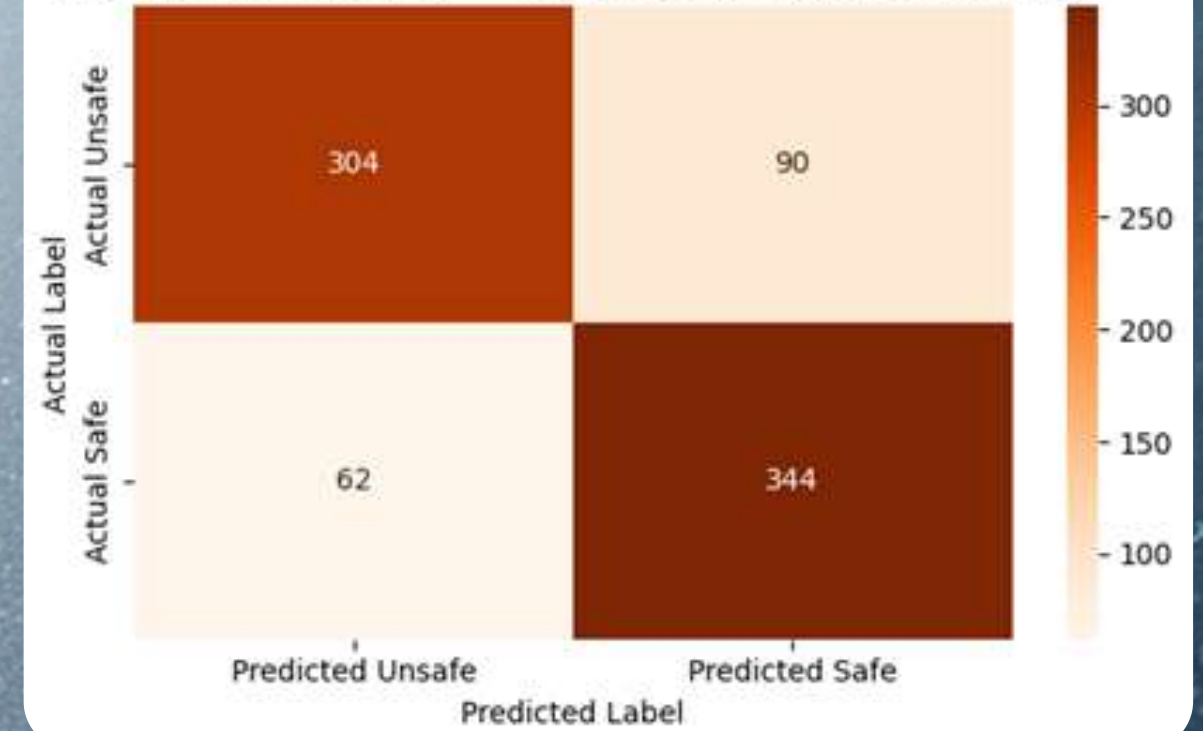
Confusion Matrix — Random Forest (Optuna Tuned)



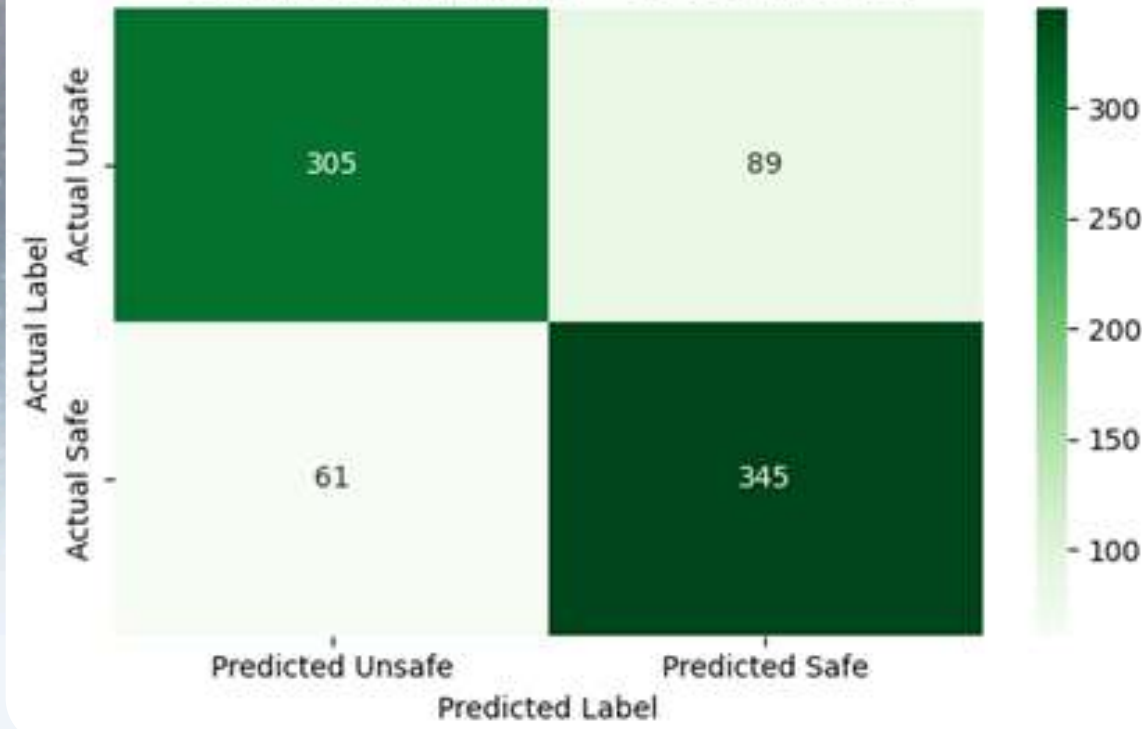
Confusion Matrix — CatBoost Model



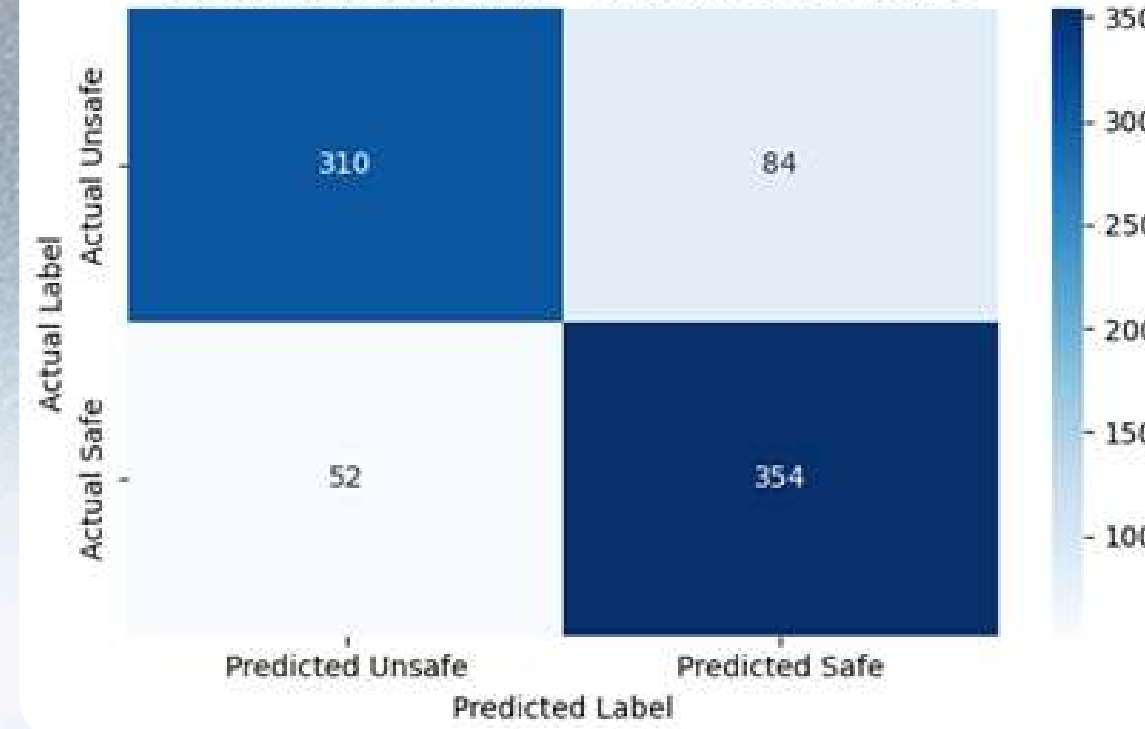
Confusion Matrix — XGBoost (Optuna Tuned)



Confusion Matrix — LGBM Model



Confusion Matrix — AdaBoost Model



CatBoost achieved the highest true positives for SAFE class, while Random Forest showed the most balanced classification overall.

Deployment Setup

- Model saved as .cbm
- Streamlit web app created
- UI + backend in app.py
- Can be deployed via FastAPI for REST API
- Requirements in requirements.txt



Streamlit UI Screenshots

Streamlit App

SafeSip

 **SafeSip – Water Safety Checker** 

Ensuring pure & safe water for every life — powered by smart ML.

 **Enter Water Parameters**

pH

7.00

-

+

Hardness

180.00

-

+

Solids

15000.00

-

+

Chloramines

7.50

-

+

Sulfate

330.00

-

+

Conductivity

500.00

-

+

Organic Carbon

10.00

-

+

Trihalomethanes

70.00

-

+

Turbidity

3.00

-

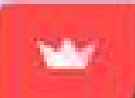
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 Predict Potability

 **The water is SAFE for Drinking — SafeSip indicates good potability.**

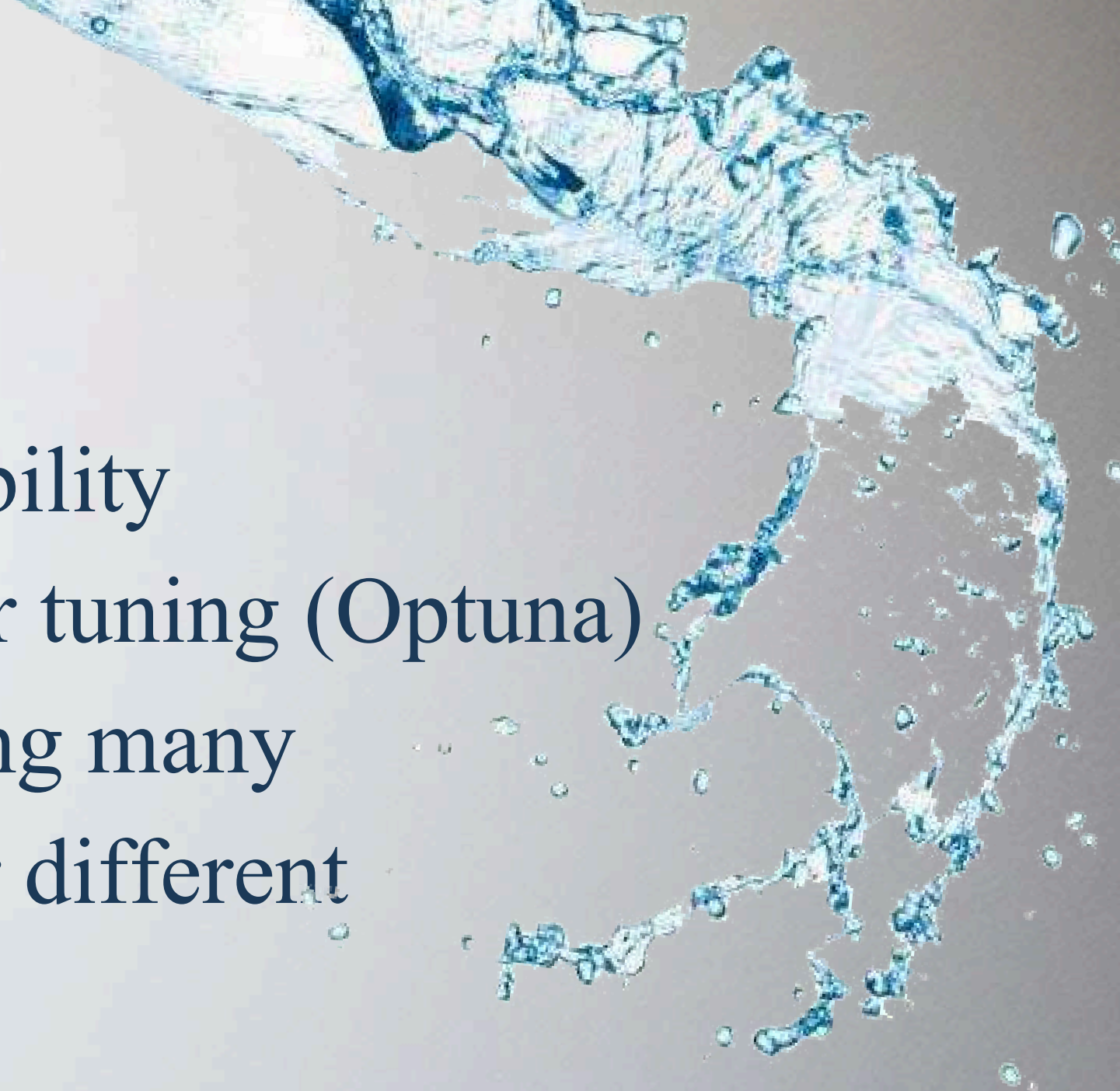
Suggestion: Continue regular monitoring. Consider filtration if taste/odor issues are observed.





Challenges Faced

- Missing and noisy values in the dataset
- Slight class imbalance affecting model stability
- Long computation time for hyperparameter tuning (Optuna)
- Difficulty in selecting the best model among many
- Handling different preprocessing needs for different algorithms
- Deployment issues like dependency management and model loading
- Time management across **EDA**, modeling, tuning, and UI design



Applications :

- Used in smart devices to check if water is safe to drink.
- Helps water cleaning plants to keep water pure.
- Useful in villages and small towns where testing labs are not available.
- Can work with IoT sensors to check water quality automatically.
- Helps government teams to find and control water pollution.
- Used in emergency or flood areas to test water quickly.
- Can be added to home water purifiers to show “Safe” or “Unsafe” water.



Conclusion

- This project helps to predict if water is safe for drinking using its chemical and physical features like pH, hardness, and turbidity.
- Among all models, **CatBoost** gave the best results with the highest accuracy.
- It can be used to detect unsafe water early and help people get clean and safe drinking water easily.



References

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- [9] Streamlit Inc., “Streamlit Documentation,” 2024. [Online].

The background is a high-quality photograph of water splashing. In the bottom left corner, a clear plastic bottle is shown pouring water, creating a dynamic splash that extends towards the top right. The water droplets are captured in mid-air, creating a sense of movement. The overall color palette is dominated by various shades of blue and teal, with some golden-brown highlights where the water reflects light. The central text 'THANK YOU' is superimposed on a bright, circular light flare that fades into the background.

THANK YOU

